

Pooled data and first differences

Slide route: Lecture10v2.tex: Starting with Pooled Data through Limits of First-Differences. Stop before Fixed Effects Models.

Session 13 | Same units or new samples?

Outcomes: Distinguish pooled cross-sections from panels, interpret time interactions, and show how first differences remove fixed unit characteristics.

100-minute route

0–10	Retrieve interaction interpretation.
10–30	Cross-section, pooled cross-section, fertility/gender examples.
30–45	Time interactions and changing associations.
45–55	Pause and identify the sampling unit.
55–75	Panel structure and traffic-fatality examples.
75–90	First differences for two and more periods.
90–100	Limits and exit exercise.

Side example | Pooled observations

Use with slides Lecture10v2 slides 5–18: Cross-Section, Pooled Cross-Section, Fertility over time, Gender Gap, and Testing Differences in Pooled regressions. Use the two-by-two interaction example while moving through slides 16–18.

In a synthetic pooled wage model,

$$wage = 10 + 2D_{\text{later}} + 3D_{\text{group}} - 1(D_{\text{later}}D_{\text{group}}) + u,$$

the group gap is 3 initially and 2 later. The interaction is the change in the gap, -1 . With new people sampled at each date, this is not the average change in the same individuals' wages.

Ask Does a two-by-two interaction automatically identify a causal difference-in-differences effect? No. That reading requires a design and additional assumptions such as a suitable parallel-trends counterfactual. Here the example is an interpretation exercise.

Panel definition A panel repeatedly observes the same units. Time labels on independent cross-sections do not turn them into a panel. Have students name the unit, time index, and outcome before interpreting any coefficient.

Worked example | Removing a fixed effect

Use with slides Lecture10v2 slides 20–34: panel data through Limits of First-Differences. Use the three-unit table when the deck reaches Traffic Fatalities first differences, especially slides 31–34.

Suppose

$$Y_{it} = \alpha_i + \beta X_{it} + \lambda_t + u_{it}.$$

With two periods,

$$\Delta Y_i = \beta \Delta X_i + \Delta \lambda + \Delta u_i.$$

The unit effect α_i cancels. A common time shock remains and can be captured by an intercept in the two-period differenced regression.

Unit	X_1	X_2	Y_1	Y_2
A	1	2	10	13
B	2	4	20	25
C	3	6	30	37

Differences are $(\Delta X, \Delta Y) = (1, 3), (2, 5), (3, 7)$. The fitted differenced relationship is

$$\Delta \hat{Y} = 1 + 2\Delta X.$$

Ask What would happen if we forced this line through the origin? The slope would be $(3 + 10 + 21)/(1 + 4 + 9) = 34/14 \simeq 2.429$. Omitting the common change contaminates this synthetic comparison.

Identification comment Removing fixed attributes does not remove time-varying confounders. If a district changes tax policy in response to an unobserved changing risk, first differences need not identify the causal effect.

More than two periods Successive differences share errors: $\Delta u_t = u_t - u_{t-1}$ and $\Delta u_{t-1} = u_{t-1} - u_{t-2}$. Even independent level errors induce dependence between neighboring differenced errors.

Exit answers Time-invariant regressors disappear under differencing. A common time shock does not disappear unless it is unchanged. Repeated independent samples cannot generally be differenced person by person.

If short / preparation Show selected years of the traffic-fatality figures rather than each in detail. Preserve the three-unit example. Next time extend removal of fixed attributes to all periods using within transformations.

Source: Lecture10v2.tex, pooled-data and first-difference sections; synthetic numerical table.